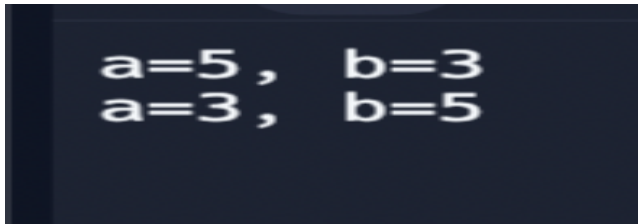


//Swap using std::swap

```
#include <iostream>
#include <utility>
using namespace std;
int main() {
    int a = 5, b = 3;
    cout << "a=" << a << ", b=" << b << endl;
    swap(a,b);
    cout << "a=" << a << ", b=" << b << endl;
    return 0;
}
```



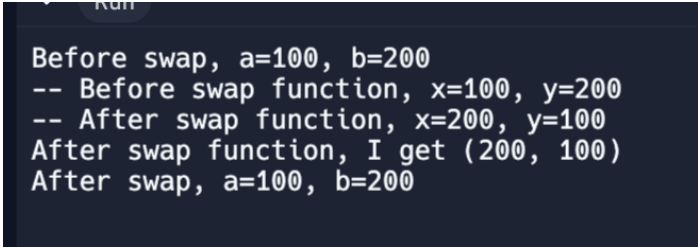
//Swap: Function Arguments: Pass By Value

```
#include <iostream>
using namespace std;

void mySwap(int x, int y) {
    cout << "-- Before swap function, x=" << x << ", y=" << y << endl;
    int temp;
    temp = x;
    x = y;
    y = temp;
    cout << "-- After swap function, x=" << x << ", y=" << y << endl;
    cout << "After swap function, I get (" << x << ", " << y << ")" << endl;
}

int main () {
    int a = 100, b=200;

    cout << "Before swap, a=" << a << ", b=" << b << endl;
    mySwap(a, b);
    cout << "After swap, a=" << a << ", b=" << b << endl;
}
```



```
Before swap, a=100, b=200
-- Before swap function, x=100, y=200
-- After swap function, x=200, y=100
After swap function, I get (200, 100)
After swap, a=100, b=200
```

//Swap: Function Arguments: Pass by Reference

```
#include <iostream>
```

```
using namespace std;
```

```
void mySwap(int &x, int &y) {
    cout << "-- Before swap function, x=" << x << ", y=" << y << endl;
    int temp;
    temp = x;
    x = y;
    y = temp;
    cout << "-- After swap function, x=" << x << ", y=" << y << endl;
}
```

```
int main () {
    int a = 100, b=200;
```

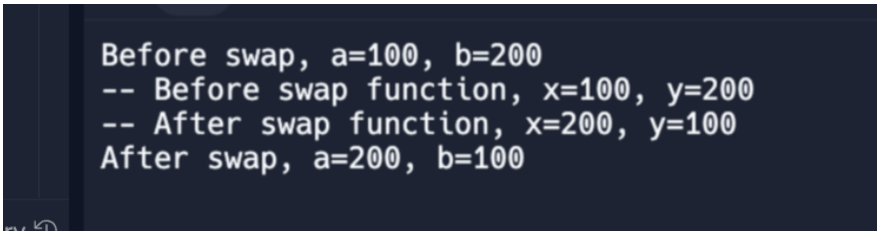
```
    cout << "Before swap, a=" << a << ", b=" << b << endl;
```

```
    mySwap(a, b);
```

```
    cout << "After swap, a=" << a << ", b=" << b << endl;
```

```
    return 0;
```

```
}
```



```
Before swap, a=100, b=200
-- Before swap function, x=100, y=200
-- After swap function, x=200, y=100
After swap, a=200, b=100
```

//Swap: Function Arguments: Pass by Pointer

```
#include <iostream>
```

```
using namespace std;
```

```
void mySwap(int* x, int* y) {
    cout << "-- Before swap function, *x=" << *x << ", *y=" << *y << endl;
    int temp;
    temp = *x;
```

```
    *x = *y;
    *y = temp;
    cout << "-- After swap function, *x=" << *x << ", *y=" << *y << endl;
}
int main () {
    int a = 100, b=200;
    cout << "Before swap, a=" << a << ", b=" << b << endl;
    mySwap(&a, &b);
    cout << "After swap, a=" << a << ", b=" << b << endl;
}
```

```
Before swap, a=100, b=200
-- Before swap function, *x=100, *y=200
-- After swap function, *x=200, *y=100
After swap, a=200, b=100
```